

B Extensions

B.1 Sensitivity to finite changes in scale

The following result formalizes the discussion in [Remark 2](#) about how the ATE for $m(Y)$ changes with finite changes in the scale of Y .

Proposition 4. *Suppose that:*

1. $m : [0, \infty) \rightarrow \mathbb{R}$ is a weakly increasing function.
2. $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$.
3. $E_{P_{Y(d)}}[|\log Y(d)| \mid Y(d) > 0] < \infty$ for $d = 0, 1$.

Then, as $a \rightarrow \infty$,

$$E_P[m(a \cdot Y(1)) - m(a \cdot Y(0))] = (P(Y(1) > 0) - P(Y(0) > 0)) \cdot \log(a) + o(\log(a)).$$

Proof. Fix a sequence $a_n \rightarrow \infty$, and without loss of generality, assume $a_n > e$. We will show that

$$\frac{1}{\log a_n} E_P[m(a_n Y(1)) - m(a_n Y(0))] \rightarrow P(Y(1) > 0) - P(Y(0) > 0). \quad (\text{B.1})$$

Define $f_n(y) = m(a_n y)/\log(a_n)$. Note that $f_n(y) \rightarrow \mathbb{1}[y > 0]$ pointwise, since $f_n(0) = m(0)/\log(a_n) \rightarrow 0$, while for $y > 0$,

$$f_n(y) = \frac{m(a_n y)}{\log(a_n)} = \frac{m(a_n y)}{\log(a_n y)} \frac{\log(a_n) + \log(y)}{\log(a_n)} \rightarrow 1,$$

where we use the fact that $m(y)/\log(y) \rightarrow 1$ as $y \rightarrow \infty$ by assumption. We apply the dominated convergence theorem to show that $E_P[f_n(Y(d))] \rightarrow P(Y(d) > 0)$.

We showed in the proof to [Proposition 1](#) that

$$|m(y)| \leq \kappa + 2 \cdot \mathbb{1}[y > 0] \cdot |\log(y)|$$

where κ is a constant not depending on y .⁴⁹ It follows that f_n is similarly dominated:

$$|f_n(y)| = \frac{|m(a_n y)|}{\log(a_n)} \leq \kappa + 2 \cdot \mathbb{1}[y > 0] \cdot (1 + |\log(y)|).$$

Further, since $E_P[|\log(Y(d))| \mid Y(d) > 0]$ is finite by assumption, the upper bound is integrable for $y = Y(d)$ for $d = 0, 1$. It follows from the dominated convergence theorem that

$$E_P[f_n(Y(d))] = E_P\left[\frac{m(a_n Y(d))}{\log(a_n)}\right] \rightarrow E_P[\mathbb{1}[Y(d) > 0]] = P(Y(d) > 0).$$

⁴⁹In particular, (A.1) implies the inequality for $\kappa = \eta + |m(0)|$.

Equation (B.1) then follows from applying this result for $d = 0, 1$ and taking the difference of the limits. $\square$

B.2 Extension to continuous treatments

Although we focus on binary treatment in the main text for simplicity, similar issues arise with continuously distributed D . Suppose now that D can take a continuum of values on some set $\mathcal{D} \subseteq \mathbb{R}$. Let $Y(d)$ denote the potential outcome at the dose d , and P the distribution of $Y(\cdot)$. Consider the parameter

$$\theta(a) = \int_{\mathcal{D}} \omega(d) E_P[m(aY(d))],$$

which is a weighted sum of the average values of $m(aY(d))$ across different values of d with weights $\omega(d)$. For example, in an RCT with a continuous treatment, a regression of $m(aY)$ on D yields a parameter of the form $\theta(a)$ where, by the Frisch–Waugh–Lovell theorem, the weights are proportional to $(d - E[D])p(d)$ and integrate to 0.⁵⁰

We now show that $\theta(a)$ can be made to have arbitrary magnitude via the choice of a when there is an extensive margin effect. In particular, by an extensive margin effect we mean that $\int \omega(d)P(Y(d) > 0) \neq 0$, i.e. when there is an average effect on the probability of a zero outcome, using the same weights $\omega(d)$ that are used for $\theta(a)$. When $\theta(a)$ is the regression of $m(aY)$ on D in an RCT, for example, $\int \omega(d)P(Y(d) > 0) \neq 0$ if the regression of $\mathbb{1}[Y > 0]$ on D yields a non-zero coefficient.

Proposition 5. *Suppose that:*

1. *The function m satisfies parts 1 and 2 of Proposition 1.*
2. *(Extensive margin effect) $\int_{\mathcal{D}} \omega(d)P(Y(d) > 0) \neq 0$.*
3. *(Bounded expectations) For all d , $E_P[|\log(Y(d))| \mid Y(d) > 0] < \infty$.*
4. *(Regularity for weights) The weights $\omega(d)$ satisfy $\int_{\mathcal{D}} \omega(d) = 0$, $\int_{\mathcal{D}} |\omega(d)| < \infty$ and $\int_{\mathcal{D}} |\omega(d)| \cdot E_P[|\log(Y(d))| \mid Y(d) > 0] < \infty$.*

Then for every $\theta^ \in (0, \infty)$, there exists $a > 0$ such $|\theta(a)| = \theta^*$. In particular, $\theta(a)$ is continuous and $\theta(a) \rightarrow 0$ as $a \rightarrow 0$ and $|\theta(a)| \rightarrow \infty$ as $a \rightarrow \infty$.*

Proof. Note that $\theta(0) = \int \omega(d)m(0) = 0$. It thus suffices to show that $\theta(a)$ is continuous for $a \in [0, \infty)$ and that $|\theta(a)| \rightarrow \infty$ as $a \rightarrow \infty$. The result then follows from the intermediate value theorem.

We first show continuity. Fix $a \in [0, \infty)$ and a sequence $a_n \rightarrow a$. Let $f_n(d) = \omega(d)E_P[m(a_n Y(d))]$. We showed in the proof to Proposition 1 that $E_P[m(a_n Y(d))] \rightarrow E_P[m(aY(d))]$, and thus $f_n(d) \rightarrow$

⁵⁰Here, $p(d)$ denotes the density of D at d over the randomization distribution.

$\omega(d)E_P[m(aY(d))]$ pointwise. We also showed in the proof to [Proposition 1](#) that for a_n sufficiently close to a ,

$$|m(a_n Y)| \leq \kappa + 2 \cdot \mathbb{1}[y > 0] \cdot |\log(y)|,$$

for a constant κ not depending on n . It follows that

$$|f_n(d)| \leq |\omega(d)| \cdot |\kappa| + 2|\omega(d)| \cdot E_P[|\log(Y(d))| \mid Y(d) > 0],$$

and the upper bound is integrable by part 4 of the Proposition. Hence, by the dominated convergence theorem, we have that $\theta(a_n) = \int_{\mathcal{D}} f_n(d) \rightarrow \int_{\mathcal{D}} \omega(d)E_P[m(aY(d))] = \theta(a)$, as needed.

To show that $|\theta(a)| \rightarrow \infty$ as $a \rightarrow \infty$, we will show that

$$\frac{\theta(a)}{\log(a)} \rightarrow \int_{\mathcal{D}} \omega(d)P[Y(d) > 0]$$

as $a \rightarrow \infty$. Consider $a_n \rightarrow \infty$, and suppose without loss of generality that $a_n > e$. Observe that

$$\frac{\theta(a_n)}{\log(a_n)} = \int_{\mathcal{D}} \omega(d) \frac{E_P[m(a_n Y(d))]}{\log(a_n)}.$$

We showed in the proof to [Proposition 4](#) that for each d ,

$$\frac{E_P[m(a_n Y(d))]}{\log(a_n)} \rightarrow P(Y(d) > 0).$$

Letting $f_n(d) = \omega(d) \frac{E_P[m(a_n Y(d))]}{\log(a_n)}$, we thus have that $f_n(d) \rightarrow \omega(d)P(Y(d) > 0)$ pointwise. Moreover, we showed in the proof to [Proposition 1](#) that

$$|m(y)| \leq \kappa + 2 \cdot \mathbb{1}[y > 0] \cdot |\log(y)|$$

where κ is a constant not depending on y . It follows that

$$\frac{|m(a_n y)|}{\log(a_n)} \leq \kappa + 2 \cdot \mathbb{1}[y > 0] \cdot (1 + |\log(y)|)$$

and thus that

$$|f_n(d)| \leq |\omega(d)| \cdot (\kappa + 2 + 2E_P[|\log(Y(d))| \mid Y(d) > 0])$$

where the upper bound is integrable by the fourth part of the proposition. The result then follows from dominated convergence. □

B.3 Extension to OLS estimands and standard errors

As noted in [Remark 4](#), our results imply that any consistent estimator of the ATE for an outcome of the form $m(aY)$ will be (asymptotically) sensitive to scaling when there is an extensive margin

effect. Our results thus cover the OLS estimator when it is consistent for the ATE for some (sub)-population P (e.g. in an RCT or under unconfoundedness). Given the prominence of OLS in applied work—and the fact that it is sometimes used for non-causal analyses—we now provide a direct result on the sensitivity to scaling of the estimand of an OLS regression of an outcome of the form $m(aY)$ on an arbitrary random variable X .

Specifically, suppose that $(X, Y) \sim Q$, for $Y \in [0, \infty)$ and $X \in \mathbb{R}^J$, where the first element of X is a constant. Consider the OLS estimand

$$\beta(a) = E_Q[XX']^{-1} E_Q[Xm(aY)],$$

i.e. the population coefficient from a regression of $m(aY)$ on X . We assume that $E_Q[XX']$ is full-rank so that $\beta(a)$ is well-defined. Letting $\beta_j(a) = e_j' \beta(a)$ be the j^{th} element of $\beta(a)$, we will show that $\beta_j(a)$ can be made to have arbitrary magnitude via the choice of a if $\gamma_j \neq 0$, where

$$\gamma = E_Q[XX']^{-1} E_Q[X \mathbb{1}[Y > 0]]$$

is the coefficient from a regression of $\mathbb{1}[Y > 0]$ on X .

Proposition 6. *Suppose that*

1. *The function m satisfies parts 1 and 2 of [Proposition 1](#).*
2. *(Finite expectations) $E_Q[\|X\|] < \infty$ and $E_Q[\|X \log(Y)\| \mid Y > 0] < \infty$.*

Then for every $j \in \{2, \dots, J\}$, $\beta_j(a)/\log(a) \rightarrow \gamma_j$ as $a \rightarrow \infty$. Moreover, if $\gamma_j \neq 0$ for some $j \in \{2, \dots, J\}$, then for every $\beta_j^ \in (0, \infty)$, there exists $a > 0$ such that $|\beta_j(a)| = \beta_j^*$. In particular $\beta_j(a)$ is continuous with $\beta_j(a) \rightarrow 0$ as $a \rightarrow 0$ and $|\beta_j(a)| \rightarrow \infty$ as $a \rightarrow \infty$.*

We note that [Proposition 6](#) implies that the OLS estimator for the j^{th} coefficient, $\hat{\beta}_j(a)$, will be arbitrarily sensitive to the choice of a when the corresponding extensive margin OLS estimator $\hat{\gamma}_j$, is non-zero. This follows immediately from setting Q to be the empirical distribution of $(Y_i, X_i)_{i=1}^N$ and applying [Proposition 6](#) (note that part 2 of the Proposition is trivially satisfied for the empirical distribution, since X and Y are both bounded over the empirical distribution).

OLS Standard Errors. We also show that as $a \rightarrow \infty$, the t -statistic for the OLS estimate $\hat{\beta}_j$ constructed using heteroskedasticity-robust standard errors converges to the t -statistic for $\hat{\gamma}_j$ (again using heteroskedasticity-robust standard errors). Formally, let

$$\hat{\Omega}_\beta(a) = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_i X_i X_i' \hat{\epsilon}_i(a)^2 \right) \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1}$$

denote the estimator of the heteroskedasticity-robust variance matrix for $\hat{\beta}(a)$, where $\hat{\epsilon}_i(a) = m(aY_i) - X_i' \hat{\beta}(a)$, and $\hat{\beta}(a)$ is the OLS estimate of $\beta(a)$. The t -statistic for $\hat{\beta}_j(a)$ is then $\hat{t}_{\beta_j}(a) =$